

Technology Stack

Date	04 July 2026
Team ID	To Be Assigned
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	2 Marks

S.No.	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side	HTML, CSS, Bootstrap	Provides a responsive and user-friendly interface for entering soil and environmental data and displaying crop recommendations.
2	Backend / Server-Side	Python, Flask	Handles user requests, processes input data, and integrates the Machine Learning model for crop prediction.
3	Database / Data Storage	CSV Dataset, Pickle Model (.pkl)	Stores the crop dataset for training and the trained Machine Learning model for prediction.
4	Cloud / Hosting / Deployment	Local Flask Server <i>(or Render if deployed)</i>	Supports application execution, testing, and future deployment.
5	Version Control & CI/CD	Git, GitHub	Enables source code management, version tracking, and team collaboration.
6	Third-Party Libraries / Tools	Scikit-learn, Pandas, NumPy, Matplotlib	Used for data preprocessing, Machine Learning model training, prediction, and data visualization.